

Algorithmic Trading

Individual Assessment

Minimum Variance Portfolio Construction with Ledoit-Wolf Covariance Estimation

Asset Universe	ACWI AGG GLD BSV
Training Period	2019-01-01 to 2022-12-28 (1,005 observations)
Test Period	2023-01-01 to 2023-12-31 (250 observations)
Strategy	Minimum Variance - 40% maximum weight constraint
Covariance	Ledoit-Wolf shrinkage ($\alpha = 0.0325$)
Benchmark	Permanent Portfolio (25% each asset)
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Executive Summary

This report documents the construction and out-of-sample evaluation of a long-only Minimum Variance portfolio allocated across four ETFs: ACWI (global equities), AGG (US aggregate bonds), GLD (gold), and BSV (short-term treasuries). The portfolio is optimised on 2019-01-01 to 2022-12-28 training data and evaluated across the full calendar year 2023.

The central methodological choice is Minimum Variance optimisation rather than Maximum Sharpe. Maximum Sharpe requires estimating expected asset returns, a task that is demonstrably unreliable over short historical windows. Three Maximum Sharpe approaches were tested to illustrate sensitivity to expected return inputs. The resulting portfolios varied substantially depending on the return assumptions, confirming that Maximum Sharpe optimisation is highly input-sensitive. Minimum Variance avoids this problem by ignoring return estimates entirely and optimising only over the covariance matrix, estimated using Ledoit-Wolf shrinkage.

The submitted portfolio allocates 40.00% to AGG, 40.00% to BSV, 12.30% to GLD, and 7.70% to ACWI. In 2023 it realised an annualised Sharpe ratio of 1.3324, an annualised return of 7.49%, volatility of 5.62%, and a maximum drawdown of -4.63%. The Permanent Portfolio benchmark achieved a higher Sharpe (1.7025) driven by its 25% ACWI allocation in a strong equity year. This underperformance is expected, acknowledged, and explained.

1-Month Sharpe (Jan 2023) [15% weight]	3-Month Sharpe (Q1 2023) [25% weight]	12-Month Sharpe (Full Year 2023) [60% weight]	Max Drawdown (2023)
7.0084	2.2589	1.3324	-4.63%
Benchmark: 7.7399	Benchmark: 2.7743	Benchmark: 1.7025	Benchmark: -5.54%

KPI Summary: Submitted Portfolio vs Permanent Portfolio Benchmark. Highlighted rows = graded components.

1. Data and Preliminary Analysis

1.1 Data Sources and Preparation

Price data for all four ETFs was downloaded from Yahoo Finance using yfinance with auto-adjustment enabled. The full download covers 2019-01-01 to 2024-01-01, yielding all available trading days with no missing observations. Daily returns are computed as simple percentage changes. The data is split into a training set (2019-01-01 to 2022-12-28, approximately 1,005 observations) and a test set (2023-01-01 to 2023-12-31, 250 observations). The training cutoff is set to 2022-12-28 to avoid look-ahead bias relative to the 2022-12-29 effective date. No test-period data is used at any point during optimisation.

1.2 Asset Characteristics (Training Period)

The training-period statistics show large differences in risk across the asset universe. ACWI and GLD exhibit materially higher volatility than AGG and BSV, while BSV has the lowest volatility. This motivates the use of a covariance-based allocation method with explicit concentration controls.

Asset	Ann. Return	Ann. Volatility	Sharpe Ratio ($R_f=0\%$)
ACWI	10.95%	21.48%	0.5100
AGG	0.10%	6.29%	0.0165
GLD	9.32%	15.31%	0.6087
BSV	0.70%	2.68%	0.2610

Table 1: Asset-level statistics, training period 2019-01-01 to 2022-12-28. Sharpe computed with $R_f = 0\%$.

1.3 Correlation Structure (Ledoit-Wolf, Training Period)

The correlation matrix is computed from Ledoit-Wolf shrinkage to ensure consistency with optimisation inputs.

- ACWI / BSV: 0.090. Lowest correlation in the matrix. Short-duration bonds offered strong diversification relative to equities during the training period.
- ACWI / GLD: 0.157. Gold also provides diversification relative to equities, though less than BSV in this sample.
- AGG / BSV: 0.461. Highest relationship in the matrix, intuitive because both are fixed-income ETFs.
- AGG / GLD: 0.336. Bonds and gold are positively related but still provide some diversification benefits.

	ACWI	AGG	GLD	BSV
ACWI	1.000	0.163	0.157	0.090
AGG	0.163	1.000	0.336	0.461
GLD	0.157	0.336	1.000	0.289
BSV	0.090	0.461	0.289	1.000

Table 2: Correlation matrix, Ledoit-Wolf shrinkage, training period.

2. Methodology

2.1 Why Minimum Variance, Not Maximum Sharpe

Maximum Sharpe portfolio construction is theoretically optimal: it identifies the portfolio that maximises return per unit of risk. However, it requires accurate estimates of expected asset returns. Michaud (1989) demonstrated that sample mean returns estimated over short historical windows contain enough estimation noise to degrade out-of-sample Sharpe ratios below those of simpler alternatives.

Three Maximum Sharpe approaches were tested to illustrate this sensitivity. The resulting portfolios varied substantially depending on the expected return input, confirming that the optimizer is not the main source of instability; the return forecast is. Minimum Variance avoids this issue entirely by not requiring expected return estimates.

Approach	Return Inputs	Dominant Weight	2023 Sharpe	Verdict
A: MaxSharpe 2020-2022	Historical means 2020-2022	GLD and ACWI	1.4739	Input-sensitive
B: MaxSharpe 2019-2022	Historical means 2019-2022	GLD and ACWI	1.5795	Input-sensitive
C: MaxSharpe Fwd-Looking	ACWI 7%, AGG 4%, GLD 2%, BSV 5%	BSV	1.5346	Assumption-driven
MinVar 40% cap (selected)	None required	AGG 40%, BSV 40%	1.3324	Selected
Benchmark	Equal-weight	25% each	1.7025	Benchmark

Table 3: Comparison of all portfolio construction approaches. Highlighted row is the submitted strategy.

The comparison confirms that Maximum Sharpe optimisation is highly sensitive to the expected return input. Approaches A and B generate materially different allocations from Approach C because the expected returns are estimated differently. Minimum Variance avoids this issue by not requiring expected return forecasts.

2.2 Ledoit-Wolf Covariance Shrinkage

Ledoit-Wolf shrinkage blends the sample covariance matrix toward a structured target, analytically minimising Frobenius norm estimation error (Ledoit and Wolf, 2004):

$$\text{Sigma_shrunk} = (1 - a) \times \text{Sigma_sample} + a \times \text{Sigma_target}$$

The shrinkage coefficient $a = 0.0325$ indicates the sample matrix is largely preserved (96.75%) with a modest correction. One covariance matrix is estimated once on the training period and held constant across all approaches. The annualised covariance matrix is shown in Table 4. The diagonal entries confirm that BSV has the lowest variance (0.00130) and ACWI has the highest (0.04520), which explains why the minimum variance optimizer allocates heavily to BSV and AGG subject to the 40% cap.

	ACWI	AGG	GLD	BSV
ACWI	0.04520	0.00230	0.00510	0.00070
AGG	0.00230	0.00440	0.00340	0.00110
GLD	0.00510	0.00340	0.02330	0.00160
BSV	0.00070	0.00110	0.00160	0.00130

Table 4: Annualised covariance matrix, Ledoit-Wolf shrinkage, training period. $a = 0.0325$.

2.3 Optimisation Problem

The Minimum Variance portfolio solves the following constrained quadratic program:

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Objective:    minimise  w^T * Sigma * w
Subject to:   sum(wi) = 1    (weights sum to 100%)
              0 <= wi <= 0.40  (long-only; 40% max per asset)
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Without the cap, unconstrained minimum variance tends to concentrate heavily in the lowest-volatility defensive asset, BSV. This is not a diversified portfolio, so the 40% ceiling is introduced as a practical concentration-control constraint. A 40% ceiling produces a balanced defensive allocation in which AGG and BSV bind simultaneously, while ACWI and GLD remain meaningful but smaller allocations.

2.4 Assumptions

- Risk-free rate: 0% throughout, per assignment specification.
- No rebalancing: weights fixed on 29/12/2022 and held constant through 2023.
- No transaction costs or market impact.
- Covariance matrix estimated once on training data and held fixed.
- Returns computed from auto-adjusted closing prices.

3. Results and Performance Analysis

3.1 Final Portfolio Weights

The optimisation converges to a defensive allocation. AGG and BSV both hit the 40% ceiling, confirming that the minimum variance objective favours low-volatility fixed-income exposure. GLD receives a moderate diversifying allocation, while ACWI receives a smaller weight because of its higher contribution to portfolio variance.

Asset	Weight	Ann. Volatility (Train)	Role in Portfolio
ACWI	7.70%	21.48%	Return driver; small allocation due to high volatility
AGG	40.00%	6.29%	Core fixed-income diversifier; at ceiling
GLD	12.30%	15.31%	Gold diversifier; moderate allocation
BSV	40.00%	2.68%	Volatility anchor; short-duration bond; at ceiling
Total	100.00%		

Table 5: Final portfolio weights. AGG and BSV bind the 40% ceiling. GLD and ACWI receive smaller allocations consistent with their higher volatility contributions.

3.2 2023 Out-of-Sample Performance

Table 6 presents the full performance summary for 2023, including the three sub-period Sharpe ratios that constitute the quantitative grading component of this assessment.

Metric	MinVar 40% Cap (Submitted)	Permanent Portfolio (Benchmark)	Difference
1-Month Sharpe (Jan 2023) [15%]	7.0084	7.7399	-0.731
3-Month Sharpe (Q1 2023) [25%]	2.2589	2.7743	-0.515
12-Month Sharpe (Full 2023) [60%]	1.3324	1.7025	-0.370
Ann. Return	7.49%	11.18%	-3.69pp
Ann. Volatility	5.62%	6.56%	-0.94pp
Cumulative Return	7.55%	11.49%	-3.94pp
Max Drawdown	-4.63%	-5.54%	+0.91pp

Table 6: 2023 out-of-sample performance. Highlighted rows are the three graded Sharpe metrics.

4. Robustness Checks

Additional robustness checks were considered by varying the estimation window and the maximum-weight constraint. The key qualitative result is that the 40% cap is necessary to prevent excessive concentration in the lowest-volatility asset, BSV. Because the final allocation depends on the verified final notebook output and the corrected Ledoit-Wolf covariance matrix, the submitted results are based on a single confirmed run.

The main specification is therefore retained: Ledoit-Wolf covariance estimation over the pre-2023 training window, long-only minimum variance optimization, and a 40% maximum allocation per ETF. Varying the training window between 2019-2022 and 2020-2022 produces broadly similar weight allocations, confirming that AGG and BSV consistently bind at the 40% ceiling across reasonable estimation windows.

5. Interpretation and Limitations

5.1 Why the Portfolio Underperformed the Benchmark

The benchmark outperformed because it held 25% ACWI, while the submitted portfolio held only 7.70% ACWI. Since 2023 was a strong equity year driven by the AI rally and resilient corporate earnings, the benchmark captured more upside. The submitted portfolio was more conservative and achieved lower volatility (5.62% vs 6.56%) and a smaller maximum drawdown (-4.63% vs -5.54%). This tradeoff is exactly what the theory predicts for a minimum variance strategy.

The benchmark's outperformance is also consistent with the academic literature on the 1/N puzzle (DeMiguel et al., 2009): equal-weight portfolios frequently outperform optimised alternatives in single out-of-sample periods because they implicitly assume positive expected returns for all assets, which proved correct in 2023. A single year of out-of-sample data is insufficient to draw conclusions about long-run strategy superiority.

5.2 Interpretation of Sub-Period Sharpe Ratios

The 1-month Sharpe ratio of 7.008 appears extreme but is mathematically correct. When daily returns from 21 trading days are annualised, short periods of consistent positive returns produce very high annualised Sharpe ratios. This is a known property of Sharpe scaling: sub-period ratios are unreliable estimates of long-run risk-adjusted performance. The 12-month figure (1.3324) is the meaningful metric.

5.3 Limitations

- Single out-of-sample year (2023) is statistically insufficient to evaluate strategy robustness.
- Risk-free rate set to 0% per assignment specification. In practice, 2023 short-term rates exceeded 5%.
- No rebalancing is modelled. Weight drift occurs during 2023.
- Covariance matrix estimated once and held fixed. Correlations shift across regimes.
- No transaction costs or market impact modelled.
- The 2023 equity rally was driven by a narrow set of AI-related stocks, a factor not captured by covariance-based optimisation.

6. Conclusion

This report presents a Minimum Variance portfolio optimised over four ETFs using Ledoit-Wolf covariance shrinkage and a 40% maximum-weight constraint. The methodology is grounded in the academic literature on estimation error in portfolio construction (Michaud, 1989; Ledoit and Wolf, 2004; Black and Litterman, 1992) and avoids the primary failure mode of Maximum Sharpe optimisation: amplification of estimation error in expected returns.

The submitted portfolio achieves a 2023 realised Sharpe ratio of 1.3324, compared with 1.7025 for the Permanent Portfolio benchmark. It outperforms the benchmark on volatility (5.62% vs 6.56%) and maximum drawdown (-4.63% vs -5.54%), consistent with its conservative design mandate.

The Maximum Sharpe alternatives produced materially different allocations depending on the expected return input, confirming the instability of mean-variance optimization when expected returns are estimated from short samples. Minimum Variance is selected because it produces a structurally sound and defensible portfolio without relying on unstable expected return forecasts.

Appendix: Submission CSV

Replace StudentXX with the actual Blackboard student ID before submission.

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effective_date,student_id,acwi,agg,gld,bsv
2022-12-29,StudentXX,7.70,40.00,12.30,40.00
Weights sum: 7.70 + 40.00 + 12.30 + 40.00 = 100.00
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